REFRIGERANT PRESSURE SENSOR [FULL-AUTO AIR CONDITIONER]

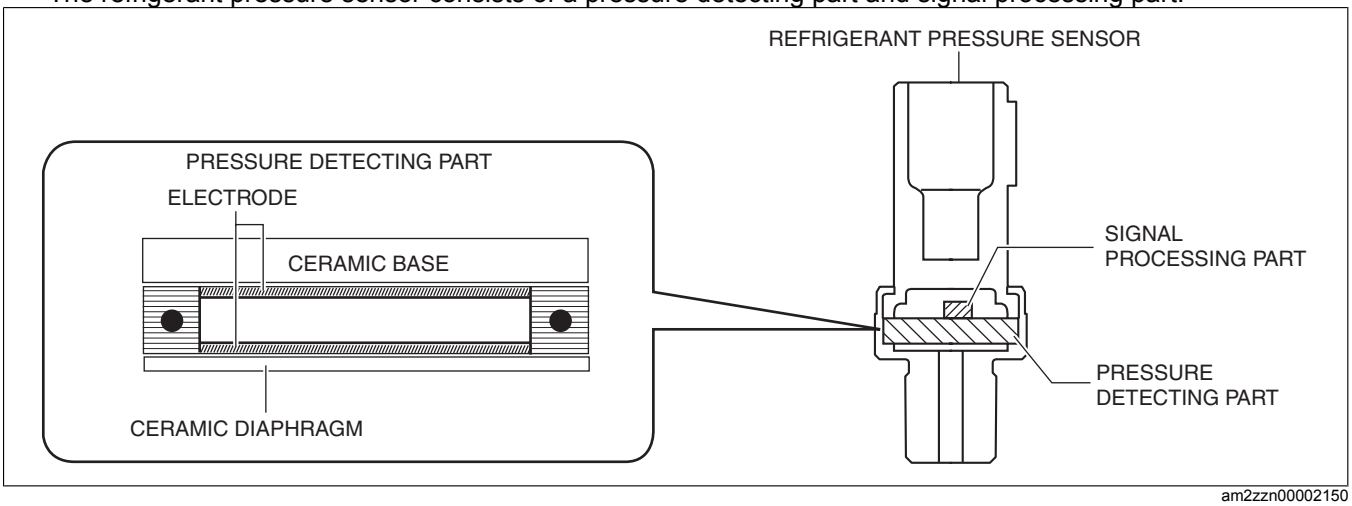
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Purpose, Function

- The refrigerant pressure sensor converts the refrigerant pressure in the detected refrigeration cycle to an electric signal.

Construction

- The refrigerant pressure sensor is located in the cooler pipe.
- The refrigerant pressure sensor consists of a pressure detecting part and signal processing part.

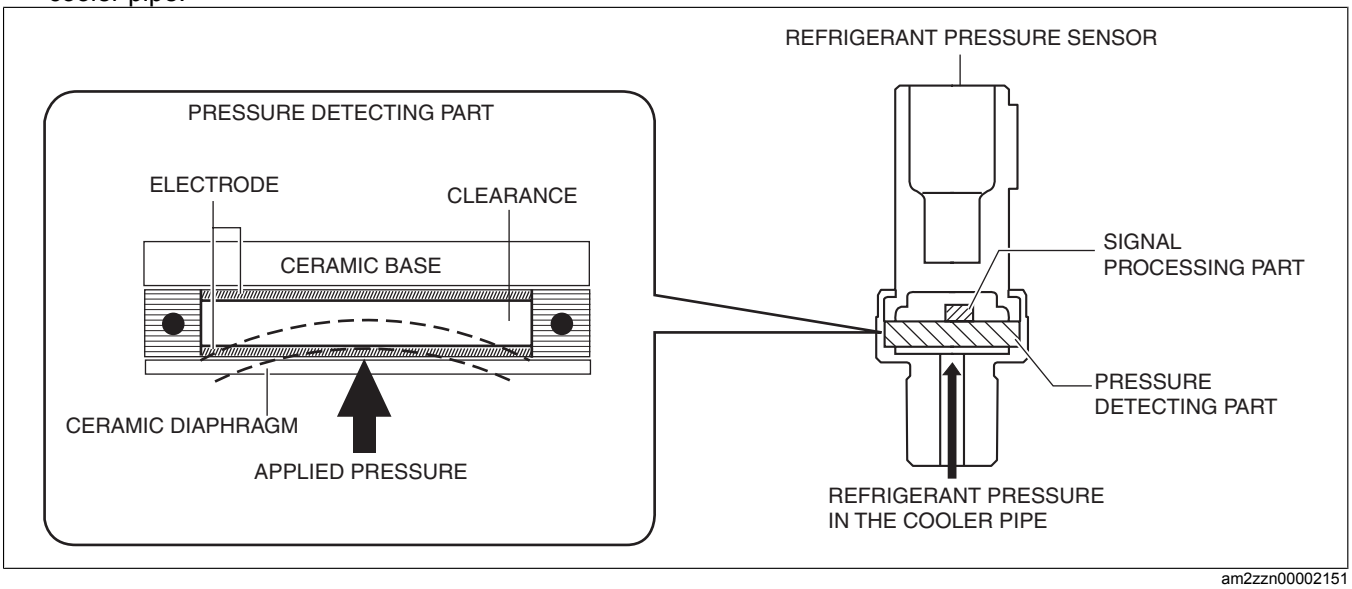


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- A capacitance type refrigerant pressure sensor has adopted to the pressure detection area.

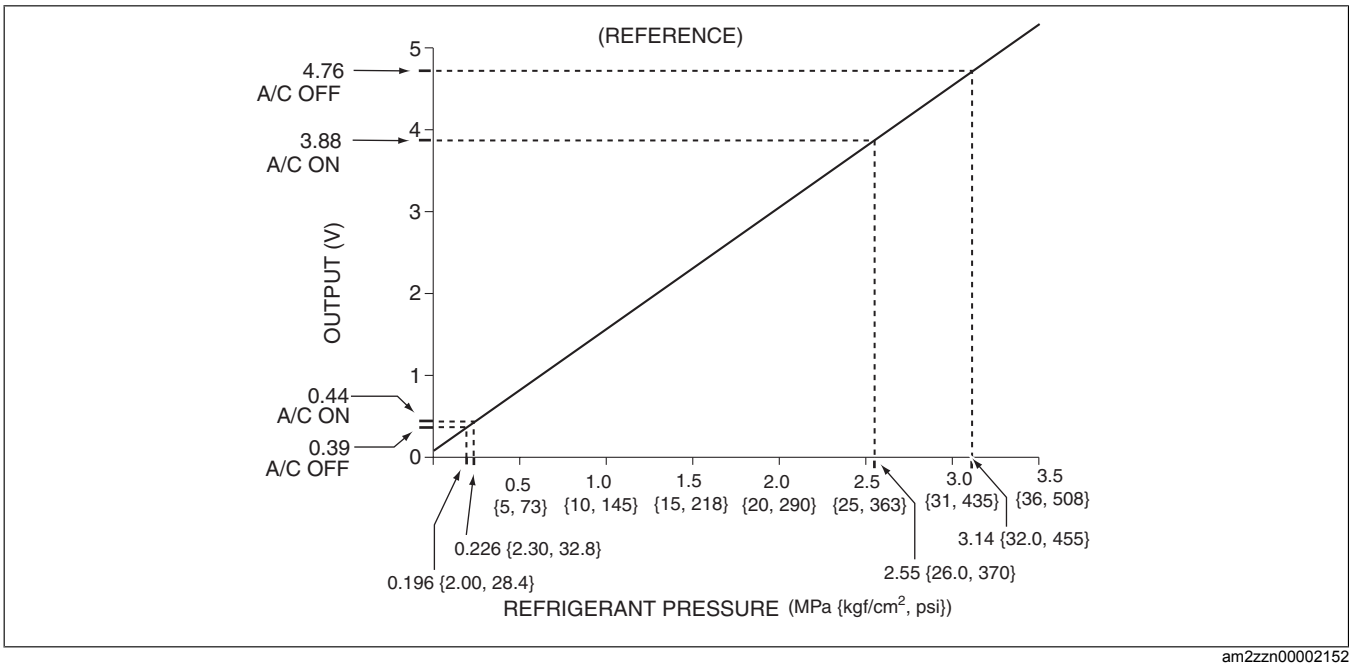
Operation

1. Pressure is applied to the ceramic diaphragm of the detection area according to the refrigerant pressure in the cooler pipe.



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2. The shape of the ceramic diaphragm changes by the applied pressure, and the amount of clearance between the electrodes changes.
3. The refrigerant pressure sensor detects the electrical load (capacitance) which changes according to the clearance amount, and converts it to a voltage value at the signal processing area. The voltage value characteristic increases if the refrigerant pressure increases, and decreases if the refrigerant pressure decreases.



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Fail-safe

- Not applicable